WATER QUALITY 19

It is important that the water supplied for home use, business and agriculture is of a good quality and free from pollutants. Some tests on water quality can be carried out in the school laboratory. One indication of water quality is the amount of other substances in the water. These materials can be **suspended** or **dissolved**.

Turbidity

Suspended material in water will make it cloudy. The small, floating particles scatter light. The amount of scattering depends on the size and concentration of the particles. Figure 1 shows two ways of measuring turbidity (apparatus not to scale). The first apparatus consists of a glass tube that has a cross on its base which can be seen by looking down from the top of the tube (not visible in figure 1). There is a scale of depth on the side of the tube. Water is poured into the open top until the cross at the base is no longer visible.

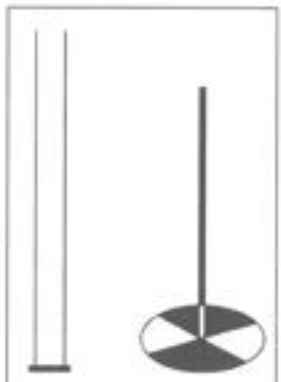


Figure 1

The second piece of equipment is a patterned disc attached to a cord which is marked at regular distances. The disc is placed into the water and lowered under water until the pattern on the disc can no longer be seen.

1 Describe how the first apparatus could be used to compare the turbidity of a range of water samples in a school laboratory.

Figure 2 shows a way to measure the amount of light transmitted through a water sample.

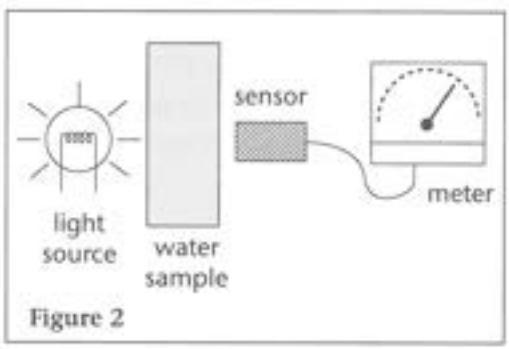


Figure 2

2 Describe how the apparatus in figure 2 could be used to compare the turbidity of water samples.

3 Discuss the accuracy of the results obtained from the apparatus in figure 1 in comparison with the apparatus in figure 2.

Total dissolved solids (TDS)

This is the total mass of solids dissolved in a known quantity. TDS is often quoted in parts per million (ppm). There are two main ways of measuring the amount of dissolved solids: evaporation and using electrical conductivity.

A student carried out a series of experiments to investigate water quality. Extracts from his report are shown below.

Evaporating water samples

The mass of an empty container was found. 100.00 mL of water sample was placed in the container and the water evaporated. The container was re-weighed. The mass of solid remaining (the residue) was calculated. This was done with five samples of water.

	Mass of empty container (g)	Mass of container + residue (g)	Mass of residue (g)
Tap water	57.51	57.76	
Bore water #1	58.36	58.69	
Bore water #2	58.02	58.82	
Sea water	56.41	59.76	
Distilled water	57.93	57.96	

Table 1

4 Name a piece of equipment that could be used to measure the volume of liquid accurately.

5 Calculate the masses of solids dissolved in the water samples.

6 State the assumptions you made in your answers to Q5.

7 Use your answers to Q5 to quote the amount of dissolved solids in the samples in ppm.

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Electrical conductivity of water samples

Ions are charged particles that can carry a current in solution. The more ions there are in solution, the greater the current that can flow. This principle can be used to measure the amount of dissolved ionic solids in a sample of water.

A student used the apparatus shown in figure 3 to investigate the conductivity of some water samples. Extracts from her report are shown below.

Method

The apparatus shown in the diagram (figure 3) was set up. The apparatus was calibrated with standard potassium chloride solutions. The conductivity of the water samples was measured.

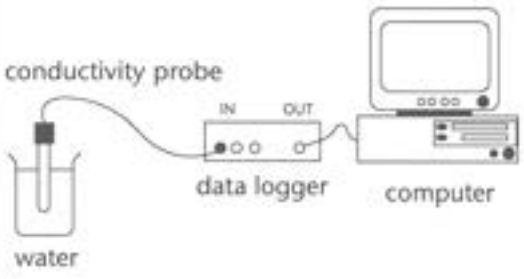


Figure 3

Results

Concentration KCl (ppm)	0	200	400	600	800	1000
Reading (arbitrary units)	20	290	550	820	1080	1350

Table 2

Sample	Distilled water	Tap water	Bore #1 water	Bore #2 water	Sea water
Reading (arbitrary units)	30	160	320	910	Off scalex

Table 3

8 Use the data from table 2 to draw a 'best-fit' line graph showing the calibration of the apparatus. Put the conductivity reading on the y-axis and the concentration of potassium chloride on the x-axis.

9 Refer to your graph from Q8 to calculate the quantity of dissolved solids in each water sample.

- **a** distilled water
- **b** tap water
- **c** bore #1 water
- **d** bore #2 water

10 Give the formula for potassium chloride.

11 Outline the purpose of the data logger in this experiment.

12 Outline the purpose of the computer in this experiment.

13 Give a possible reason for why the reading for sea water was off the scale.

14 Outline the assumptions made in this method of finding total dissolved solids in a sample of water.

15 What is meant by *arbitrary units*?

16 What is meant by *calibrate*?

17 What is meant by *turbidity*?

18 Discuss the advantages and disadvantages of using electrical conductivity compared to evaporation of water samples to find the amount of total dissolved solids in a sample of water.

Research

19 Define *nephelometer*.

20 Find out the accepted TDS levels for *potable* (fit for human consumption) water in your area.

21 Chlorine is a poisonous substance yet it is commonly used in drinking water and swimming pools. Why is it used? Why doesn't it poison humans?

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